

Replay-Aware ATSP Aggregate Report

Overview

- Run count: 20.
- Condition count: 4.
- Best mean transfer gap: atsp_failure_replay_compression.

Condition Comparison

Condition	Mean Transfer Gap	Mean TSPLIB ATSP Gap	Mean Synthetic Gap	Mean Accepted Novelty	Mean Complexity	Mean Adaptation Efficiency
atsp_failure_replay	0.103224	0.188856	0.017592	0.567311	0.660375	0.014901
atsp_failure_replay_compression	0.102242	0.187225	0.017259	0.617612	0.66	0.017559
atsp_no_replay	0.108551	0.192847	0.024256	0.36888	0.699875	0.011466
atsp_random_replay	0.103757	0.192742	0.014773	0.565185	0.64	0.014107

Bootstrap Confidence Intervals

atsp_failure_replay

- Transfer gap 95% CI: 0.098679 to 0.10785.
- TSPLIB ATSP gap 95% CI: 0.180163 to 0.195733.
- Synthetic gap 95% CI: 0.012457 to 0.02328.
- Mean accepted novelty 95% CI: 0.399203 to 0.724716.

atsp_failure_replay_compression

- Transfer gap 95% CI: 0.097779 to 0.107925.
- TSPLIB ATSP gap 95% CI: 0.177656 to 0.197249.
- Synthetic gap 95% CI: 0.012663 to 0.022272.
- Mean accepted novelty 95% CI: 0.449587 to 0.764254.

atsp_no_replay

- Transfer gap 95% CI: 0.102438 to 0.115582.
- TSPLIB ATSP gap 95% CI: 0.182104 to 0.204743.
- Synthetic gap 95% CI: 0.018273 to 0.030237.
- Mean accepted novelty 95% CI: 0.199128 to 0.54588.

atsp_random_replay

- Transfer gap 95% CI: 0.099275 to 0.109755.
- TSPLIB ATSP gap 95% CI: 0.186304 to 0.201261.

- Synthetic gap 95% CI: 0.010363 to 0.02.
- Mean accepted novelty 95% CI: 0.393863 to 0.723033.

Paired Comparisons

Negative delta favors the candidate for transfer gap, benchmark gap, synthetic gap, novelty, and complexity. Positive delta favors the candidate for adaptation efficiency.

Comparison	Paired Runs	Transfer Delta	Transfer 95% CI	ATSP Delta	ATSP 95% CI	Novelty Delta	Complexity Delta	Adaptation Delta
atsp_random_replay vs atsp_no_replay	20	-0.004794	-0.014338 to 0.004763	-0.00105	-0.014844 to 0.014854	0.196305	-0.059875	0.002641
atsp_failure_replay vs atsp_no_replay	20	-0.005327	-0.012952 to 0.001875	-0.003991	-0.016859 to 0.008555	0.198431	-0.0395	0.003435
atsp_failure_replay vs atsp_random_replay	20	-0.000533	-0.008292 to 0.006606	-0.003886	-0.01533 to 0.0059	0.002126	0.020375	0.000794
atsp_failure_replay_compression vs atsp_failure_replay	20	-0.000982	-0.007357 to 0.00623	-0.001631	-0.014403 to 0.011915	0.050301	-0.000375	0.002658
atsp_failure_replay_compression vs atsp_no_replay	20	-0.006309	-0.0133 to -0.000173	-0.005622	-0.019126 to 0.007143	0.248732	-0.039875	0.006093

Paired Notes

- atsp_random_replay vs atsp_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.004794 with 95% CI -0.014338 to 0.004763; candidate better on 65.0% of paired runs.
- Mean TSPLIB ATSP-gap delta: -0.000105 with 95% CI -0.014844 to 0.014854.
- atsp_failure_replay vs atsp_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.005327 with 95% CI -0.012952 to 0.001875; candidate better on 70.0% of paired runs.
- Mean TSPLIB ATSP-gap delta: -0.003991 with 95% CI -0.016859 to 0.008555.
- atsp_failure_replay vs atsp_random_replay used 20 paired runs.
- Mean transfer-gap delta: -0.000533 with 95% CI -0.008292 to 0.006606; candidate better on 50.0% of paired runs.
- Mean TSPLIB ATSP-gap delta: -0.003886 with 95% CI -0.01533 to 0.0059.
- atsp_failure_replay_compression vs atsp_failure_replay used 20 paired runs.
- Mean transfer-gap delta: -0.000982 with 95% CI -0.007357 to 0.00623; candidate better on 50.0% of paired runs.
- Mean TSPLIB ATSP-gap delta: -0.001631 with 95% CI -0.014403 to 0.011915.
- atsp_failure_replay_compression vs atsp_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.006309 with 95% CI -0.0133 to -0.000173; candidate better on 45.0% of paired runs.
- Mean TSPLIB ATSP-gap delta: -0.005622 with 95% CI -0.019126 to 0.007143.